

SESSION 16

Django Integrations

Topics Covered: Social Login (Google), Email APIs (Mailchimp / Mailgun), SMS APIs (Twilio)

Activities: Integrate Google Login for patient, Send OTP via SMS

Task 1: Integrate Google Login into Django project so users sign in using Google account, display name/email on homepage.

Explanation & Approach: Uses *social-auth-app-django* package with *OAuth 2.0 Client Credentials*.

```
# settings.py
INSTALLED_APPS = [
    # ...
    'social_django',
]

AUTHENTICATION_BACKENDS = (
    'social_core.backends.google.GoogleOAuth2',
    'django.contrib.auth.backends.ModelBackend',
)

SOCIAL_AUTH_GOOGLE_OAUTH2_KEY = 'your-client-id.apps.googleusercontent.com'
SOCIAL_AUTH_GOOGLE_OAUTH2_SECRET = 'your-client-secret'

LOGIN_URL = 'login'
LOGOUT_URL = 'logout'
LOGIN_REDIRECT_URL = 'home'

# urls.py
urlpatterns = [
    path('oauth/', include('social_django.urls', namespace='social')),
]
```

Task 2: Set up email sending feature in Django using Mailgun API — when user registers, send welcome email with username using requests library.

Explanation & Approach: Calls *Mailgun REST API* endpoint via *Python requests library*.

```
import requests

def send_mailgun_welcome(user_email, username):
    api_key = "YOUR_MAILGUN_API_KEY"
    domain = "YOUR_MAILGUN_DOMAIN"

    return requests.post(
        f"https://api.mailgun.net/v3/{domain}/messages",
        auth=("api", api_key),
        data={
            "from": f"Welcome Team <mailgun@{domain}>",
```

```

        "to": [user_email],
        "subject": f"Welcome aboard, {username}!",
        "text": f"Hi {username},\n\nThank you for signing up on our platform!"
    }
)

```

Task 3: Build Django view that sends 6-digit OTP to mobile number using Twilio SMS API and displays confirmation message.

Explanation & Approach: Uses official Twilio REST Client SDK to dispatch SMS messages to user handsets.

```

from twilio.rest import Client
import random
from django.shortcuts import render, HttpResponse

def send_twilio_otp(request):
    account_sid = 'YOUR_TWILIO_ACCOUNT_SID'
    auth_token = 'YOUR_TWILIO_AUTH_TOKEN'
    twilio_number = '+1234567890'

    user_phone = '+919876543210' # Target user mobile number
    otp = str(random.randint(100000, 999999))

    client = Client(account_sid, auth_token)
    message = client.messages.create(
        body=f"Your Verification OTP is: {otp}",
        from_=twilio_number,
        to=user_phone
    )

    return HttpResponse(f"OTP sent successfully via Twilio! SID: {message.sid}")

```

Task 4: Use ChatGPT/Copilot to generate Django model and view for storing and verifying OTP codes sent via SMS, then adapt and review.

Explanation & Approach: Presents AI-assisted OTP model schema with timestamp expiration checks.

```

# AI-Drafted & Verified OTP Model & Logic
from django.db import models
from django.utils import timezone
import datetime

class SMSOTPVerification(models.Model):
    phone_number = models.CharField(max_length=15)
    otp_code = models.CharField(max_length=6)
    created_at = models.DateTimeField(auto_now_add=True)
    is_verified = models.BooleanField(default=False)

    def is_valid(self):
        # Expiry set to 3 minutes (180 seconds)

```

```
        now = timezone.now()
        return not self.is_verified and (now - self.created_at) <
datetime.timedelta(minutes=3)
```